

Evaluating Emerging Autonomous Vehicle (AV) Congestion, Safety, and Modal Shift Dynamics Using SFCTA's Activity-Based (SF-CHAMP) and Dynamic Network Simulation Framework

The integration of autonomous vehicles (AVs) into dense urban environments like San Francisco necessitates a paradigm shift in how transportation authorities regulate safety, evaluate modal shifts, and measure accessibility. As AV operators transition from supervised testing to driverless commercial deployments, the San Francisco County Transportation Authority (SFCTA) leverages its advanced San Francisco Chained Activity Modeling Process (SF-CHAMP) alongside dynamic, programmatic spatial pipelines to measure system impacts¹. This report details the methodological approaches utilized to deconstruct these emerging dynamics, comparing conventional practices with the advanced programmatic frameworks proposed in the AV_SFCTA_Policy_Project GitHub plan.

1. Conventional vs. Proposed Methodologies: A Comparative Review

Evaluating transportation networks traditionally relied heavily on macroscopic models and graphical user interface (GUI) Geographic Information Systems (GIS). Modern frameworks demand continuous, automated programmatic architectures.

Conventional Spatial and Statistical Methods

Traditionally, spatial modeling relied on proprietary desktop GIS environments (e.g., ArcGIS) where analysts manually executed spatial joins, buffering, and topological corrections⁴. Data structures like Shapefiles were standard but lacked topological awareness⁵. Statistically, conventional travel demand forecasting relied on aggregate, trip-based models. These models evaluated single movements from an origin to a destination in isolation¹. To calculate consumer surplus (economic benefit) from policy changes, analysts routinely used the "Rule of Half," an aggregate measure heavily dependent on macroscopic demand curves that struggles to account for complex multi-modal shifts⁶.

Limitations: These manual workflows become severe bottlenecks when evaluating high-frequency AV telemetry or real-time transit delays. Traditional trip-based models fail to capture the interdependent nature of human travel (e.g., driving an AV to work limits midday transit options).

The Proposed GitHub Project Methodology

The project plan dictates a structural shift to open-source spatial data science pipelines and

activity-based modeling.

- **Spatial Modeling Deviations:** The project replaces manual GIS tasks with programmatic libraries like GeoPandas, Shapely, and PostGIS for automated, reproducible spatial filtering and continuous database indexing⁴. It utilizes OSMnx to construct algorithmically corrected spatial street network graphs (NetworkX directed multigraphs), which mathematically merge complex nodes and allow for immediate network routing updates⁴.
- **Statistical Modeling Deviations:** The SFCTA utilizes SF-CHAMP, a tour-based, activity-based model that evaluates continuous chains of trips rather than isolated movements¹. It utilizes Random Utility Maximization (RUM) and discrete choice theory to output disaggregate logsums, replacing the Rule of Half to better capture socio-demographic heterogeneity¹⁰.

Rationale and Limitations: This programmatic architecture aligns with modern data science standards, ensuring algorithm transparency and continuous integration of high-velocity data. However, the limitation of this approach is its computational overhead and steep learning curve for analysts trained in traditional GUI environments. Furthermore, while the logsum mathematically aggregates accessibility, its outputs lack intuitive interpretability for policymakers compared to simple travel time metrics¹².

2. Statistical and Spatial Modeling Primer: Utility Curves and Logsums

To evaluate modal shift dynamics (e.g., a commuter choosing an AV over Muni rail), the framework relies on discrete choice models driven by utility functions.

The Utility Function and Probability

The utility U_{nj} that individual n derives from transport alternative j contains an observable deterministic component (V_{nj}) and a random stochastic error (ϵ_{nj}) assumed to follow a Gumbel distribution¹²:

$$U_{nj} = V_{nj} + \epsilon_{nj}$$

The standard form of the deterministic utility is typically a linear combination of coefficients and variables:

$$V_{nj} = \beta_0 + \beta_1 \cdot \text{Cost}_{nj} + \beta_2 \cdot \text{InVehicleTime}_{nj}$$

When introducing AVs, the project adds terms to capture unique behavioral dynamics:

$$V_{n,AV} = \beta_0 + \beta_1 \cdot \text{Cost} + \beta_2 \cdot \text{Time} + \beta_3 \cdot \text{AV_Tech_Affinity} + \beta_4 \cdot \text{Unplanned_}$$

The probability P_{nj} of choosing alternative j is derived using the standard logistic function:

$$P_{nj} = \frac{e^{V_{nj}}}{\sum_{i \in J} e^{V_{ni}}}$$

Interpreting 'y' and Mock Graphical Demonstrations

In standard choice modeling, the y-axis represents P_{nj} , the probability of choosing a specific mode (bounded between 0 and 1). The x-axis represents the Generalized Cost (a monetized combination of time and out-of-pocket expenses).

Expected Behavior & Model Shifts:

1. **Shift 1 (Standard Transit vs. Base AV):** The AV curve sits further to the right because users may have a higher base utility (β_0) for privacy. They will tolerate a higher generalized cost before their probability of choosing the AV drops.
2. **Shift 2 (Increasing Unplanned Delay Penalty):** If the AV project data reveals a high rate of AV immobilizations, the β_4 penalty increases. This mathematically reduces $V_{n,AV}$, shifting the S-curve sharply to the left. The user will abandon the AV mode at much lower cost thresholds.
3. **Shift 3 (AV Tech Affinity):** A positive β_3 coefficient for younger demographics flattens the upper tail of the curve, representing sustained mode loyalty despite minor cost increases.

Mock Graph: Probability of Mode Choice vs. Generalized Cost

P_{nj} (Probability of Choosing Mode)

1.0 |----- <-- Shift 3: AV w/ High Tech Affinity

|-----

0.8 | \ ----- <-- Shift 1: Base AV Ride-hail

| \

0.5 | \ \

| \

0.2 | Shift 2: \ \

| AV w/ High \

0.0 | **Delay** _____ Generalized Cost (\$) \$10 \$20 \$30 \$40 *Interpretation of y (P_{nj}):*

A y-value of 0.8 indicates an 80% likelihood that an individual with specific socio-demographics will choose this mode at the corresponding cost.

The Logsum Model (Consumer Surplus)

To measure network-wide equity and congestion impacts, the denominator of the logit probability is extracted. The logsum calculates the expected maximum utility (converted to

consumer surplus $E(CS_n)$ by dividing by the marginal utility of income (α_n) across the entire

choice set J ⁶:

$$E(CS_n) = \frac{1}{\alpha_n} \ln \left(\sum_{j \in J} e^{V_{nj}} \right) + C$$

Interpretation of y ($\text{Logsum} / E(CS_n)$): Here, the y -value is a monetized measure of societal benefit.

- **Broad Expectations in Results:** Introducing a new AV service mathematically adds an alternative to choice set J , expanding the sum of exponentials and increasing the logsum (positive consumer surplus). However, if dynamic network simulations reveal that AV cruising significantly increases congestion, the V_{nj} for all other modes (buses, personal cars) will drop due to higher travel times. If the negative utility impact on public transit outweighs the private benefit of the AVs, the overall logsum will decrease, signaling a net societal accessibility loss¹⁰.

3. Granular Data Cleaning and Quality Assurance Methodology

Fueling these spatial networks and discrete choice models requires merging highly fragmented, noisy data streams. The project plan establishes rigorous automated cleaning rules that draw heavy structural parallels to healthcare claims adjudication—where overlapping, distinct reports of the same clinical incident must be collapsed into a single "episode of care" using deterministic deduplication keys.

Data Source Benchmarking and QA Matrix

Data Source	Granularity & Refresh	Raw Variables	Derived Variables	Deduplication & Merging Keys	QA & Cleaning Methods
NHTSA SGO API (AV Crashes) [cite: 14, 15, 16]	Event-level. Initial report + 10-day strict updates.	Crash_Time stamp, VIN, Pre_Crash_Speed, Injury_Severity,	Automation_Category (Level 2 vs 3-5), Episode_ID.	VIN exact match AND Timestamp ± 5 mins AND Spatial radius	Spatial clustering to collapse duplicate multi-entity reports. Rule-based

		Consumer_Narrative.		(50m).	heuristic re-classification of Level 2 vs Level 3-5 capabilities ¹⁰ .
GTFS-Realtime (SFMTA/511) [cite: 17, 18, 19]	Point-location. Sub-minute API polling.	Vehicle_ID, Coordinates, Stop_Sequence_ID, Service_Alert_Text.	$\text{Delay}_{AV,Net}$ (Delay vector), Actual_Travel_Time.	Vehicle_ID AND Trip_ID over continuous time-series index.	"Early skip" algorithms to halt trajectory tracking. Imputation of missed stops. Dropping GPS anomalies (e.g., velocity > or > jumps) ²⁰ .
SF OpenData (HIN & EPCs) [cite: 21, 22]	Street segment (HIN), Census Block Group (EPC). 5-year rolling / Annual ACS.	Segment_ID, Crash_Severity, Minority_Pct, Low_Income_Pct.	High_Injury_Flag, EPC_Status.	Spatial Intersects (EPSG:26910 NAD83 UTM Zone 10N).	Spatial projection transformations. Snapping raw AV crash points to the nearest algorithmically corrected OSMnx street edge ⁴ .

Deduplication and Harmonization Techniques

1. **NHTSA SGO Deduplication:** Similar to an outsourced anesthesiologist and a hospital filing two claims for one surgery, a technology developer and a ride-hail operator may

legally file reports for the exact same AV crash under the SGO 30-second rule¹⁵. The project plan utilizes a composite key linking VIN, temporal proximity (± 5 mins), and spatial boundaries to mathematically drop duplicates.

2. **GTFS-RT Telemetry QA:** GTFS-RT protocol buffers are fundamentally transient; once a vehicle passes a stop, the real-time update is overwritten¹⁸. The pipeline employs continuous ingestion to build a persistent data warehouse. Because transit vehicles do not strictly halt at every scheduled coordinate, spatial algorithms interpolate trajectories and calculate the true $\text{Delay}_{\text{AV,Net}}$ vector by contrasting the real-time geographic breadcrumbs against static schedules¹⁷.
3. **Spatial Harmonization:** All ingested API data frequently arrives in varied coordinate reference systems (CRS) or lacking them entirely. The project plan enforces a strict QA rule (Rule 2) that imputes missing CRS data and programmatically transforms all geographic data into a unified local projection (NAD83 UTM Zone 10N) using GeoPandas, ensuring that distance calculations for transit delays and spatial joins with Equity Priority Communities are metrically sound.

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